ПРАКТИКУМ З ПРОГРАМУВАННЯ №5

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```
=== Main Menu ===
1 - Create account
2 - Login
3 - Exit
Your choice: 1
Enter username: kirilo
Enter password (must contain at least 1 digit and 1 of: $, %, _): 1%$
Enter phone number (10 digits, starting with 05/06/07/09): 0671234567
Are you a broker?
1 - Yes
2 - No
Your choice: 2
Account created successfully!

=== Main Menu ===
1 - Create account
2 - Login
3 - Exit
Your choice: 2
Enter username: kirilo
Enter password: 1%$
Welcome, kirilo!

=== Menu ===
1 - Post new property
2 - Remove listing
3 - View all listings
4 - My listings
5 - Search
6 - Logout
Your choice: 1
Available cities:
1 - Kyiv
2 - Lviv
3 - Dnipro
4 - Kharkiv
Enter city name: 2
City not found.

=== Menu ===
1 - Post new property
2 - Remove listing
3 - View all listings
4 - My listings
5 - Search
6 - Logout
Your choice: 1
Available cities:
1 - Kyiv
2 - Lviv
3 - Dnipro
4 - Kharkiv
Enter city name: Lviv
Streets in Lviv:
  - Franka
  - Horodotska
  - Sakharova
Enter street name: Franka
Property type:
1 - Regular apartment
2 - Penthouse
3 - Private house
Your choice: 3
Number of rooms: 2
Building number: 54
1 - For sale
2 - For rent
```

```
Your choice: 1
Price (UAH): 24500
Listing posted successfully!

=== Menu ===
1 - Post new property
2 - Remove listing
3 - View all listings
4 - My listings
5 - Search
6 - Logout
Your choice: 3
1. Private house for sale: 2 room(s).
Address: Lviv, Franka st., bld. 54
Price: 24500 UAH
Contact: kirilo, 0671234567.
---

=== Menu ===
1 - Post new property
2 - Remove listing
3 - View all listings
4 - My listings
5 - Search
6 - Logout
Your choice: 5
Sale or rent? (1 - sale, 2 - rent, 999 - any): 32
Property type (1 - apartment, 2 - penthouse, 3 - house, 999 - any): 52
Number of rooms (999 - any): 23
Minimum price (999 - ignore): 53
Maximum price (999 - ignore): 23
No results found.

=== Menu ===
1 - Post new property
2 - Remove listing
3 - View all listings
4 - My listings
5 - Search
6 - Logout
Your choice: 6

=== Main Menu ===
1 - Create account
2 - Login
3 - Exit
Your choice: 
```

```
#define _USE_MATH_DEFINES

#include <iostream>

#include <string>
```

```
#include <limits>

using namespace std;

const int MAX_USERS = 100;
const int MAX_PROPERTIES = 200;
const int MAX_ADDRESSES = 10;

struct Address {
    string city;
    string street;

    void toString() {
        cout << "Street: " << street << ", City: " << city << endl;
    }
};

struct User {
    string username;
    string password;
    string phone;
    bool isBroker;
    bool isNull;

    void toString() {
        cout << username << ", " << phone;
        if (isBroker) cout << " (broker)";
        else cout << " (regular user)";
        cout << endl;
    }
};
```

```
struct Property {
    Address address;
    int rooms;
    double price;
    int type;
    bool forSale;
    int buildingNumber;
    int floor;
    User owner;
    bool isNull;

    void toString() {
        if (type == 1) cout << "Regular apartment";
        else if (type == 2) cout << "Penthouse";
        else cout << "Private house";

        if (forSale) cout << " for sale: ";
        else cout << " for rent: ";

        cout << rooms << " room(s)";
        if (type == 1 || type == 2) cout << ", floor " << floor;
        cout << "." << endl;

        cout << "Address: " << address.city << ", " << address.street << " st., bld. " << buildingNumber
<< endl;
        cout << "Price: " << price << " UAH" << endl;
        cout << "Contact: " << owner.username << ", " << owner.phone;
        if (owner.isBroker) cout << " (broker)";
        cout << "." << endl;
    }
};
```

```
Address addresses[MAX_ADDRESSES];
User users[MAX_USERS];
Property properties[MAX_PROPERTIES];
int userCount = 0;
int propertyCount = 0;

void initAddresses() {
    addresses[0] = { "Kyiv", "Shevchenko" };
    addresses[1] = { "Kyiv", "Khreshchatyk" };
    addresses[2] = { "Kyiv", "Lesi Ukrainky" };
    addresses[3] = { "Kyiv", "Velyka Vasylkivska" };
    addresses[4] = { "Lviv", "Franka" };
    addresses[5] = { "Lviv", "Horodotska" };
    addresses[6] = { "Lviv", "Sakharova" };
    addresses[7] = { "Dnipro", "Soborna" };
    addresses[8] = { "Dnipro", "Robocha" };
    addresses[9] = { "Kharkiv", "Sumska" };
}

bool isValidPassword(const string& pass) {
    bool hasDigit = false;
    bool hasSpecial = false;
    for (char c : pass) {
        if (isdigit(c)) hasDigit = true;
        if (c == '$' || c == '%' || c == '_') hasSpecial = true;
    }
    return hasDigit && hasSpecial;
}

bool isValidPhone(const string& phone) {
    if (phone.length() != 10) return false;
    for (char c : phone)
```

```
        if (!isdigit(c)) return false;
    string prefix = phone.substr(0, 2);
    return (prefix == "05" || prefix == "06" || prefix == "07" || prefix == "09");
}

bool usernameExists(const string& name) {
    for (int i = 0; i < userCount; i++)
        if (users[i].username == name) return true;
    return false;
}

void createUser() {
    User newUser;
    newUser.isNull = false;

    string username;
    do {
        cout << "Enter username: ";
        cin >> username;
        if (usernameExists(username))
            cout << "Username already taken. Try another." << endl;
    } while (usernameExists(username));
    newUser.username = username;

    string password;
    do {
        cout << "Enter password (must contain at least 1 digit and 1 of: $, %, _): ";
        cin >> password;
        if (!isValidPassword(password))
            cout << "Password does not meet requirements. Try again." << endl;
    } while (!isValidPassword(password));
    newUser.password = password;
```

```
    string phone;
    do {
        cout << "Enter phone number (10 digits, starting with 05/06/07/09): ";
        cin >> phone;
        if (!isValidPhone(phone))
            cout << "Invalid phone number. Try again." << endl;
    } while (!isValidPhone(phone));
    newUser.phone = phone;

    int brokerChoice;
    cout << "Are you a broker?" << endl;
    cout << "1 - Yes" << endl;
    cout << "2 - No" << endl;
    cout << "Your choice: ";
    cin >> brokerChoice;
    newUser.isBroker = (brokerChoice == 1);

    users[userCount++] = newUser;
    cout << "Account created successfully!" << endl;
}

User userLogin() {
    User nullUser;
    nullUser.isNull = true;

    string username, password;
    cout << "Enter username: ";
    cin >> username;
    cout << "Enter password: ";
    cin >> password;
```

```
    for (int i = 0; i < userCount; i++)
        if (users[i].username == username && users[i].password == password)
            return users[i];

    return nullUser;
}

int countUserProperties(const User& user) {
    int count = 0;
    for (int i = 0; i < propertyCount; i++)
        if (properties[i].owner.username == user.username) count++;
    return count;
}

void printCities() {
    string shownCities[MAX_ADDRESSES];
    int shownCount = 0;

    for (int i = 0; i < MAX_ADDRESSES; i++) {
        bool found = false;
        for (int j = 0; j < shownCount; j++)
            if (shownCities[j] == addresses[i].city) { found = true; break; }
        if (!found) shownCities[shownCount++] = addresses[i].city;
    }

    cout << "Available cities:" << endl;
    for (int i = 0; i < shownCount; i++)
        cout << i + 1 << " - " << shownCities[i] << endl;
}

bool postNewProperty(User user) {
    int limit = user.isBroker ? 10 : 3;
```

```
if (countUserProperties(user) >= limit) {
    cout << "You have reached the listing limit (" << limit << ")." << endl;
    return false;
}

string shownCities[MAX_ADDRESSES];
int cityCount = 0;
for (int i = 0; i < MAX_ADDRESSES; i++) {
    bool found = false;
    for (int j = 0; j < cityCount; j++)
        if (shownCities[j] == addresses[i].city) { found = true; break; }
    if (!found) shownCities[cityCount++] = addresses[i].city;
}

cout << "Available cities:" << endl;
for (int i = 0; i < cityCount; i++)
    cout << i + 1 << " - " << shownCities[i] << endl;

int cityChoice;
cout << "Enter city number: ";
cin >> cityChoice;

if (cityChoice < 1 || cityChoice > cityCount) {
    cout << "Invalid city number." << endl;
    return false;
}
string chosenCity = shownCities[cityChoice - 1];

string shownStreets[MAX_ADDRESSES];
int streetCount = 0;
for (int i = 0; i < MAX_ADDRESSES; i++)
    if (addresses[i].city == chosenCity)
```

```
        shownStreets[streetCount++] = addresses[i].street;

    cout << "Streets in " << chosenCity << ":" << endl;
    for (int i = 0; i < streetCount; i++)
        cout << i + 1 << " - " << shownStreets[i] << endl;

    int streetChoice;
    cout << "Enter street number: ";
    cin >> streetChoice;

    if (streetChoice < 1 || streetChoice > streetCount) {
        cout << "Invalid street number." << endl;
        return false;
    }
    string chosenStreet = shownStreets[streetChoice - 1];

    Address chosenAddress;
    bool streetFound = false;
    for (int i = 0; i < MAX_ADDRESSES; i++) {
        if (addresses[i].city == chosenCity && addresses[i].street == chosenStreet) {
            chosenAddress = addresses[i];
            streetFound = true;
            break;
        }
    }

    if (!streetFound) {
        cout << "Street not found." << endl;
        return false;
    }

    Property newProp;
```

```
newProp.isNull = false;
newProp.address = chosenAddress;
newProp.owner = user;

cout << "Property type:" << endl;
cout << "1 - Regular apartment" << endl;
cout << "2 - Penthouse" << endl;
cout << "3 - Private house" << endl;
cout << "Your choice: ";
cin >> newProp.type;

if (newProp.type == 1 || newProp.type == 2) {
   cout << "Floor number: ";
   cin >> newProp.floor;
}
else {
   newProp.floor = 0;
}

cout << "Number of rooms: ";
cin >> newProp.rooms;

cout << "Building number: ";
cin >> newProp.buildingNumber;

int saleChoice;
cout << "1 - For sale" << endl;
cout << "2 - For rent" << endl;
cout << "Your choice: ";
cin >> saleChoice;
newProp.forSale = (saleChoice == 1);
```

```
    cout << "Price (UAH): ";
    cin >> newProp.price;

    properties[propertyCount++] = newProp;
    cout << "Listing posted successfully!" << endl;
    return true;
}

void removeProperty(User user) {
    int indices[MAX_PROPERTIES];
    int count = 0;

    for (int i = 0; i < propertyCount; i++)
        if (properties[i].owner.username == user.username)
            indices[count++] = i;

    if (count == 0) {
        cout << "You have no listings." << endl;
        return;
    }

    cout << "Your listings:" << endl;
    for (int i = 0; i < count; i++) {
        cout << i + 1 << ". ";
        properties[indices[i]].toString();
        cout << "---" << endl;
    }

    int choice;
    cout << "Enter listing number to remove: ";
    cin >> choice;
```

```
    if (choice < 1 || choice > count) {
        cout << "Invalid number." << endl;
        return;
    }

    int removeIdx = indices[choice - 1];
    for (int i = removeIdx; i < propertyCount - 1; i++)
        properties[i] = properties[i + 1];
    propertyCount--;

    cout << "Listing removed successfully." << endl;
}

void printAllProperties() {
    if (propertyCount == 0) {
        cout << "No listings available." << endl;
        return;
    }
    for (int i = 0; i < propertyCount; i++) {
        cout << i + 1 << ". ";
        properties[i].toString();
        cout << "---" << endl;
    }
}

void printUserProperties(const User& user) {
    int count = 0;
    for (int i = 0; i < propertyCount; i++) {
        if (properties[i].owner.username == user.username) {
            cout << ++count << ". ";
            properties[i].toString();
            cout << "---" << endl;
```

```
        }
    }
    if (count == 0) cout << "You have no listings." << endl;
}

void search() {
    int saleFilter, typeFilter, roomsFilter;
    double minPrice, maxPrice;

    cout << "Sale or rent? (1 - sale, 2 - rent, 999 - any): ";
    cin >> saleFilter;
    cout << "Property type (1 - apartment, 2 - penthouse, 3 - house, 999 - any): ";
    cin >> typeFilter;
    cout << "Number of rooms (999 - any): ";
    cin >> roomsFilter;
    cout << "Minimum price (999 - ignore): ";
    cin >> minPrice;
    cout << "Maximum price (999 - ignore): ";
    cin >> maxPrice;

    int found = 0;
    for (int i = 0; i < propertyCount; i++) {
        Property& p = properties[i];
        if (saleFilter != 999 && ((saleFilter == 1) != p.forSale)) continue;
        if (typeFilter != 999 && p.type != typeFilter) continue;
        if (roomsFilter != 999 && p.rooms != roomsFilter) continue;
        if (minPrice != 999 && p.price < minPrice) continue;
        if (maxPrice != 999 && p.price > maxPrice) continue;

        cout << ++found << ". ";
        p.toString();
        cout << "---" << endl;
```

```
    }
    if (found == 0) cout << "No results found." << endl;
}

void innerMenu(User user) {
    int choice;
    do {
        cout << endl << "=== Menu ===" << endl;
        cout << "1 - Post new property" << endl;
        cout << "2 - Remove listing" << endl;
        cout << "3 - View all listings" << endl;
        cout << "4 - My listings" << endl;
        cout << "5 - Search" << endl;
        cout << "6 - Logout" << endl;
        cout << "Your choice: ";
        cin >> choice;

        switch (choice) {
        case 1: postNewProperty(user); break;
        case 2: removeProperty(user); break;
        case 3: printAllProperties(); break;
        case 4: printUserProperties(user); break;
        case 5: search(); break;
        case 6: break;
        default: cout << "Invalid choice." << endl;
        }
    } while (choice != 6);
}

void mainMenu() {
    int choice;
    do {
```

```
        cout << endl << "=== Main Menu ===" << endl;
        cout << "1 - Create account" << endl;
        cout << "2 - Login" << endl;
        cout << "3 - Exit" << endl;
        cout << "Your choice: ";
        cin >> choice;

        if (choice == 1) {
            createUser();
        }
        else if (choice == 2) {
            User logged = userLogin();
            if (logged.isNull) {
                cout << "Invalid username or password." << endl;
            }
            else {
                cout << "Welcome, " << logged.username << "!" << endl;
                innerMenu(logged);
            }
        }
        else if (choice == 3) {
            cout << "Goodbye!" << endl;
        }
        else {
            cout << "Invalid choice." << endl;
        }
    } while (choice != 3);
}

int main() {
    initAddresses();
    mainMenu();
```

```
    return 0;
}
```

У ході виконання практичної роботи було розроблено консольний застосунок на мові C++, який реалізує систему оголошень про нерухомість. Програма підтримує реєстрацію та вхід користувачів, розміщення та видалення оголошень, пошук за параметрами. Реалізовано валідацію всіх вхідних даних та розмежування прав між брокерами та звичайними користувачами. Набуто навичок роботи зі структурами, масивами та функціями у мові C++.